

Phase 1 build brief

Phase 1: detection, surface, statistics and the application, build brief

Written 14 September 2026 by the planning session **Follows** docs/PHASE0-RESULTS.md and docs/NOTES-FROM-PLANNING.md entry 11 **Paper** none of this brief needs paper. The one paper session is next weekend and has its own protocol document

0. Read these first, in this order

- docs/PHASE0-RESULTS.md, all of it, and in particular section 4.5 and the decision log in section 7. Phase 1 inherits four things and they are listed in section 8.
- docs/NOTES-FROM-PLANNING.md entry 11, which is the geometry commit and the frozen-fixture rule.
- DESIGN.md sections 6, 13 and 21. Section 6 already names the surface model. Section 13 requires an explicit assignment interface. Section 21 carries the gates.
- docs/SCAN-MEASUREMENTS.md section 3, and section 8 for the naive baselines. Section 3.1 is the hole-detection primitive that already works, and Phase 1 starts from it rather than from nothing.
- docs/STATISTICS.md, all of it. Section 15 is a build specification, not a sketch.
- docs/FIDUCIAL-DECISION.md section 10, measurement 2, which is the marker module sweep.

1. What Phase 1 is for

Phase 0 proved the sheet can be registered from its own printed fiducials on a scan and on a flat photograph. Phase 1 turns that into an application: it registers a sheet that is not flat, finds the holes, computes the statistics, and puts all three behind a window a person can use.

Four milestones, in this order, for reasons given under each. **Report at the end of each milestone and continue to the next without waiting**, except at M1, where the report changes a document that has a deadline.

Milestone	What	Paper	Why here
M1	Developable surface fit	None. The nine pinned frames are committed	Highest unknown in the project, and its result decides what the paper protocol asks for
M2	Hole detection	None. 343 holes are committed, plus synthetic	Needs nothing from M1 but follows it because M1 has a deadline
M3	Statistics engine	None at all, ever	Fully specified, fully independent, and the Monte Carlo tables want hours of idle compute
M4	Application shell	None	Where the other three meet. Last because it needs them

M0, a small thing to do first, in an hour. The marker module sweep of FIDUCIAL-DECISION.md section 10 measurement 2 needs five sheets at 0.3, 0.4, 0.5, 0.6 and 0.8 mm modules. fiducials.markerSize is already a declared field with validation against family, and tag36h11 is ten modules across, so this is five definitions at markerSize 30, 40, 50, 60 and 80 dmm, generated by the layout tool, each with its own identifier. **No renderer change and no schema change.** Emit them as a sweep set, let the validator say what it says about the 30 dmm sheet rather than suppressing it, and report the PDF page count so the protocol knows what it is asking to be printed. They are not built-in library sheets; keep them under a measurement path.

2. The gates

Unchanged from DESIGN.md section 21 as amended [r5], plus two that are new to Phase 1.

Gate	Measured on	Threshold
Conformance test 43	Synthetic raster	0.001 in worst bull. Must not regress

Paper gate	The ten printed sheets	0.005 in worst bull. Must not regress
Photograph gate, flat	<code>main_flat1-3</code>	0.005 in worst bull. The control for M1
Photograph gate, mounted	The seven usable pinned frames	0.005 in worst bull. M1's gate
Hole detection	See section 4.4	Stated there, because it is new
Statistics	<code>STATISTICS.md</code> section 15.5	That section, unchanged. It was DESIGN's Phase 2 gate and is being pulled forward

3. M1: the developable surface fit

3.1 The model, and why it is this one

`DESIGN.md` section 6 states it and `PHASE0-RESULTS.md` section 4.5 argues it: **paper bends without stretching**. A sheet of paper has zero Gaussian curvature everywhere, so the map from page coordinates to the sheet in space is an isometry: distance measured along the sheet is the same distance it was on the flat page. That is a constraint, and Phase 0's evidence is that a constraint is what is missing. Three radial coefficients with a free centre, and quadratic and cubic warps evaluated leaving one marker out, all failed, and every one of them gave the fit *more* freedom.

Start with a generalised cylinder, which is the simplest developable surface and the one a mounted target actually makes. The sheet bends about a family of parallel straight lines, the rulings. A target stapled along two edges and bowing between them is a generalised cylinder. A target with one free edge curling forward is a generalised cylinder. The parameters are:

- **One angle**, the direction of the rulings in page coordinates.
- **A bend profile**, the shape of the cross-section perpendicular to the rulings, as a function of arc length along the page. Three to five coefficients, a cubic or a small spline. Arc length, not page distance projected, is the point: the isometry is what makes the profile a function of arc length at all.
- **Six camera extrinsics**, rotation and translation.
- **The lens**, k_1 and k_2 , already fitted in Phase 0 and consistent across frames at k_1 -0.044 to -0.073, k_2 +0.053 to +0.106.

Call it fifteen parameters against 136 corners, which is 272 equations on a full-marker frame and 182 on the worst. Comfortably over-determined, which is the other reason to start here.

If a frame will not fit a generalised cylinder, the next step is a general developable surface with non-parallel rulings, a cone or a tangent-developable, which a sheet pinned at a single point can genuinely make. Do not skip to it. Fit the cylinder first, report which frames it fails, and let the failures say what the next parameterisation needs to be. A twist requires it; a simple bow does not.

Do not build an as-rigid-as-possible mesh. It is the right answer to a harder problem and it will absorb error that belongs to the detector.

3.2 The trap: you need a focal length now, and you did not before

A homography with a radial term works without knowing anything about the camera, because a plane maps to a plane under projection and the unknowns fold into the eight homography coefficients. **A non-planar surface does not have that property.** Fitting a shape in three dimensions and projecting it requires real perspective, which requires the focal length in pixels.

EXIF gives it. The pinned frames carry a focal length and a 35 mm equivalent, the main camera reporting 6.25 mm at 23 mm equivalent on a 4000 by 3000 frame, which puts the initial estimate at about 4000 times 23 divided by 36, near 2556 pixels. Use that as a starting value and refine it as a parameter, one per camera rather than one per frame where the EXIF says the frames share a lens. Report the refined value against the EXIF value: if they disagree badly, the EXIF or the fit is wrong and it matters which.

`photos.json` already carries `focalLengthMm`, `fNumber` and `focalLength35mm` per frame. Frames with no focal length tag cannot use this path at all, which is a real product limitation and belongs in the report.

3.3 Build it against synthetic truth first, and touch the real frames once

This is the method Phase 0 used for the bull locator and for the lens, and it is why those results are trustworthy. The decision log's line "far-edge marker loss reported, not tuned against" is the same principle, and it is the single most important instruction in this brief.

1. **Render bent sheets with known truth.** `Rendering/SceneRasterizer.cs` already rasterises a scene. Extend the path so a page can be mapped through a known generalised cylinder and a known camera before rasterising: known ruling angle, known bend profile, known extrinsics, known focal length, and the measured lens coefficients. Add corner noise matching what Phase 0 measured, not an invented figure.
2. **Recover the truth.** Check that the fit returns the bend it was given and, more importantly, the bull positions. Report bull error against truth, not parameter error against truth: a wrong ruling angle that lands the bulls correctly is not a failure.
3. **Sweep until it breaks.** Increase bend amplitude, add a twist the model cannot represent, cut the marker count to the 23 that `main_flat3` decoded, move the camera off-axis to the 0.802 keystone of the worst real frame. Report where the 0.005 in gate is crossed in each direction. That sweep is the result, more than a pass or a fail on nine frames.
4. **Then run the nine pinned frames once, and report.** Whatever comes back, report it. Do not adjust the model and re-run to improve the number. If the first run suggests a change, make the change on synthetic evidence and say in the report that the pinned figure is from the second run and why.

3.4 The control, which is as important as the gate

The three flat frames must not get worse. A flat sheet is a degenerate generalised cylinder with zero bend, so the fit must recover approximately zero bend on `main_flat1-3` and reproduce, to within noise, the homography-and-lens result of `PHASE0-RESULTS.md` section 3b: worst scoring bulls 0.00343, 0.00566 and 0.01183 in, and on `main_flat1` all 25 scoring bulls inside the gate. A surface model that improves the pinned frames by bending the flat ones is overfitting and the control is how you see it.

Run the scans through it too. The ten printed sheets on a flatbed are as flat as anything gets, and the paper gate must not regress from 0.00325 in worst.

3.5 What to report

The benchmark and the figures to beat are in `PHASE0-RESULTS.md` sections 3a and 4.5: worst scoring bull 0.015 to 0.091 in across the seven usable pinned frames, 8 to 21 of 25 scoring bulls over the gate, corner RMS 0.014 to 0.060 in over all corners.

Report per frame: recovered ruling angle, bend amplitude in inches of deflection, refined focal length, corners kept and total, residual over kept and over all, bull mean, worst scoring bull, worst sighter, scoring bulls over the gate, and whether it needed a cylinder or something more general. Raw rows as `scans/phase0/measurements/surface.json` under the entry 7 convention. Compare against the whole-sheet homography and against the nearest-marker diagnostic of section 4.5 in the same table, so the three approaches sit side by side.

Two things will still fail and they are not the model's fault. A bull outside the marker lattice is extrapolated whatever the surface model, which is section 4.4 reached from a fourth direction, though a developable fit does constrain the surface past the last marker in a way a piecewise fit cannot, and that difference is worth measuring explicitly. And markers lost to defocus are a detection problem, listed as Phase 1's second inheritance and not solved here.

3.6 If it fails

Say so plainly and report the sweep from 3.3, which is what tells us whether the model is wrong or the frames are worse than the model's range. The fallback is the piecewise registration of section 4.5, whose diagnostic figures are already measured. **Do not quietly become the fallback.** If the constrained fit cannot be made to work, that is a result that changes the product, and it changes what the paper protocol asks for next weekend, which is why M1 is first.

4. M2: hole detection

4.1 Start from what already works

`docs/SCAN-MEASUREMENTS.md` section 3.1 already found the primitive, and section 8 already showed what the obvious approaches do. On `300_nm_hand_load.jpg`, against 27 hand-verified holes:

Method	True positives	False positives
Filled contour, threshold and circularity swept	1 of 27 at best	up to 5, all footer text
Gradient energy	7 of 20	
Neutral darkness, opened, closed, filled, convex hull	25 of 27	0

The primitive:

```
Dn      = paper_level - max(R, G, B)
hole_map = open(Dn, disk r = 0.032 in) -> threshold -> close(disk r = 0.055 in) -> fill -> convex hull
```

It works for reasons the survey states and that are worth keeping in the code comments: the opening disk at 0.064 in is wider than the widest printed stroke anywhere in the corpus, which is a 0.0567 in barcode bar against a 0.0525 in ring stroke, so opening erases printed artwork and cannot erase a hole. Neutral darkness near-zeroes red, pink and blue artwork because those keep one channel at paper level. The convex hull matters because the rim is frequently a C rather than an O.

Port it into GroupLab.Core and revalidate it on all 343 holes across the fifteen files, plus the messaging-app copy at its own DPI. That is the baseline, and it is committed before anything is built to beat it.

4.2 Then render-and-difference, which is what the fiducials are for

A GroupLab sheet is registered, and the application knows exactly what was printed on it. So it can render the sheet as it should look, map it through the recovered registration, and difference. Nothing in the existing corpus can test this, because none of those targets is a GroupLab target and none has a known definition.

So this week it is tested on synthetic sheets, and next weekend it is tested on real ones.

4.3 Synthetic holes, which the survey already characterised well enough to build

This is the part that makes M2 possible without paper, and the survey did the work without knowing it would be used this way. `SCAN-MEASUREMENTS.md` gives the core intensity distribution in section 3.3, the angle-averaged radial intensity profile in section 3.4, hole diameter against nominal bullet diameter in section 3.5, and the difference between a hole on printed ink and a hole on bare paper in section 3.6.

Punch synthetic holes into a rendered GroupLab sheet, drawn from those measured distributions, with known centres. Then both methods can be measured against truth, on known geometry, at any hole count. Encode the hard cases the survey found as fixtures, because every one of them is real and already documented:

- **The bright core is the scanner lid**, not the hole. Synthetic scans have a white backing and synthetic photographs do not, and section 3.1 of the survey plus `SAMPLE-NOTES.md` both say so. Get this wrong and the synthetic set teaches the detector something false.
- **Overlapping perforations merge into one blob**, as at bull 20 of `300_nm_hand_load`.
- **X marks over holes** do not remove the hole but inflate the hull by 0.03 to 0.07 in.
- **Arrowheads and the closed bowls of letters** are compact, dark and hole-sized after closing. Five bowls in one blue caption were detected as holes.
- **Holes on printed ink versus bare paper** behave differently, per section 3.6.
- **A hole partly off the edge of a bull**, and a hole that destroys part of a ring, which is the bull the naive detector missed in `SAMPLE-NOTES.md`.

4.4 The gate

New, so it is stated here rather than inherited.

1. **Against the ported baseline on real paper**: on the fifteen scanned files, no worse than the survey's own figures, and on `300_nm_hand_load` at least 25 of 27 with zero false positives. A regression here means the port is wrong.
2. **Against truth on synthetic GroupLab sheets**: render-and-difference recovers at least 99 percent of holes with no false positives at a hole-centre tolerance of 0.01 in, at hole counts up to two per bull, and degrades gracefully rather than catastrophically as registration error grows. Sweep registration error deliberately and report where it breaks, because that is what couples M2 to M1.
3. **Hole centre accuracy against truth** reported, not gated, at this stage. The 0.008 in centroid noise floor measured

on real paper is the number it will eventually be judged against, and a synthetic figure far below it means the synthesis is too clean.

4.5 Assignment

SCAN-MEASUREMENTS.md section 7.2 settles the rule and it is cheap: where the definition declares one shot per bull, the defensible assignment is the globally optimal one-to-one matching, Hungarian, minimising total distance. Nearest-bull is wrong on exactly the case where one cell holds two shots and a neighbour holds none. Implement the matching, keep nearest-bull as a comparison, and report where they differ.

DESIGN.md section 13 requires an explicit assignment interface regardless, because the survey found hand-drawn arrows recording the shooter's own cross-cell assignment, which no geometric rule recovers. That interface is M4's.

5. M3: the statistics engine

docs/STATISTICS.md is the specification and section 15 is the build plan, with fixtures in 15.2, tolerances in 15.3, the nine shotGroups quirks to encode as expected in 15.4, and the gate in 15.5. Implement it as written. This was DESIGN's Phase 2 gate and is being pulled forward because it needs nothing from anyone.

The two open questions in section 16 are answered here, so they do not block you.

1. **Apply the c4 bias correction to the confidence-interval endpoints, matching shotGroups**, and document in the reference material that the interval therefore does not have exactly nominal coverage for sigma. Your own reasoning for matching is right: divergence would put a special case in every validation comparison, and the divergence is small and explicable where a permanent special case is neither.
2. **Mean radius with its interval is the headline**, sigma immediately beneath as the underlying estimate, extreme spread present and visually subordinate. Mean radius is a fixed multiple of sigma, so nothing is lost statistically, and it is the figure the audience already reads. Extreme spread stays because leaving it out would make the application look wrong to every shooter who opens it, and it is subordinate because it is the worst estimator in the document. That is a product decision and it is mine to make.

The Monte Carlo tables are the reason this milestone suits this week. Section 15.3 wants at least ten million replications per cell for n from 2 to 50 and groups from 1 to 10, which is hours of compute in a separate tool, run once, with the output checked in. Start that running early and let it work while you build something else. Acceptance is within 0.2 percent on the mean and 0.5 percent on the 2.5 and 97.5 percent quantiles against shotGroups' own table.

Licence care, per section 15.1. shotGroups is GPL-2-or-later and GroupLab is GPL-3.0, so linking is not the issue, but GroupLab must not ship shotGroups' data or its Monte Carlo tables as its own. Generated fixtures under test/fixtures/shotgroups/ with a README stating provenance, package version, R version and licence, and attribution to Wollschlaeger in the repository per DESIGN.md section 4.

6. M4: the application shell

Avalonia, GPL-3.0, and the point is that Alan can open it and click on things. Minimum useful scope, in order:

1. Open an image. Show it.
2. Register it, showing which markers were found and which were not. Stage S3 of the trace already says of every missing marker whether it was found and unread or never found; surface that.
3. Overlay the recovered bulls and the detected holes on the image.
4. **Correct an assignment by hand**, which is DESIGN.md section 13 and is not optional. Drag a hole to a different bull. Mark a detection as not a shot. Mark a shot as excluded.
5. Show the statistics panel for the group, with mean radius as the headline per section 5 above.
6. Export the result.

Comment the code for a reader who did not write it. That has been the standing requirement since Phase 0a and it matters more here, because the UI is where an outside contributor is most likely to start.

Do not build a settings system, a plugin architecture, or a theme engine. Do not add analysis the statistics document does not specify.

7. What not to do

- **Do not tune anything against the frames or scans it is gated on.** Build against synthetic truth, evaluate once, report. This is the rule Phase 0 followed and it is why Phase 0's numbers are worth something.
- **Do not touch the frozen Phase 0 definitions** once entry 11's fixtures exist. They are inputs to a measurement.
- **Do not change the gates.** If evidence says a gate is wrong, that is a question for the planning session, in `docs/QUESTIONS-FOR-PLANNING.md`, with the evidence attached.
- **Do not add pseudoscience.** Barrel harmonics, optimal barrel time, velocity nodes and accuracy nodes are not real and must not appear in code, comments, UI copy or documentation. If a statistic looks like it is heading toward one of those claims, it is out of scope.
- **Do not implement any OnTarget compatibility,** read any OnTarget format, or reproduce any OnTarget target design.
- **Do not request paper.** The one paper session is next weekend and its protocol is being written separately. Anything you want from it goes in `docs/QUESTIONS-FOR-PLANNING.md` as soon as you know, because the protocol has a print deadline and a late addition costs a whole week.

8. Reporting

As Phase 0: `docs/PHASE1-RESULTS.md`, one document, sections per milestone, every table reproducible by a named command, raw rows under `scans/phase0/measurements/` or a Phase 1 equivalent per entry 7, and a decision log of one line per method choice where there was a real alternative. That decision log was the most useful part of the Phase 0 report and it should grow the same way.

Report at the end of each milestone and keep going. The exception is M1: its report is wanted as soon as it exists, because the paper protocol for next weekend depends on what the surface fit turns out to need, and that protocol has to be in Alan's hands with time to print.

If something blocks you, it goes in `docs/QUESTIONS-FOR-PLANNING.md` with its evidence, and you stop. The mailbox convention is in `CONTRIBUTING.md` and it has answered five questions in five round trips so far.